Task 1: Optimizer Performance on Non-Convex Functions

1. Objectives

- Implement the Rosenbrock and simplified 1D Ackley functions and their gradients.
- Implement Gradient Descent, SGD with Momentum, RMSprop, Adagrad and Adam from scratch.
- Compare the optimizers at learning rates 0.01, 0.05 and 0.1.
- Study convergence behaviour and record the final objective value, iterations and time.
- Check whether the experimental behaviour agrees with the expected effect of learning rate and optimizer design.
- Use the results to identify which methods performed best for the two functions.

2. Functions Used

2.1 Rosenbrock Function

The function used was:

$$f(x, y) = (1 - x)^2 + 100(y - x^2)^2$$

Its global minimum is at $(x, y) = (1, 1)$, where $f(x, y) = 0$. The starting point used for the main experiment was $(-1.5, 2.0)$.

2.2 Simplified 1D Ackley Function

$$f(x) = -20 \exp(-0.2\sqrt{(x^2)}) - \exp(\cos(2\pi x)) + 20 + e$$

2.3 Optimization Algorithms

Optimizer	Working principle
Gradient Descent	Updates parameters directly in the opposite direction of the current gradient.
Momentum	Maintains a velocity term that accumulates gradients and can accelerate movement along consistent directions.
Adam	Uses first- and second-moment estimates with bias correction, giving parameter-wise adaptive step sizes.
RMSprop	Scales the gradient using a moving average of squared gradients.
Adagrad	Accumulates squared gradients to adapt the effective learning rate over time.

3. **Experimental Methodology**

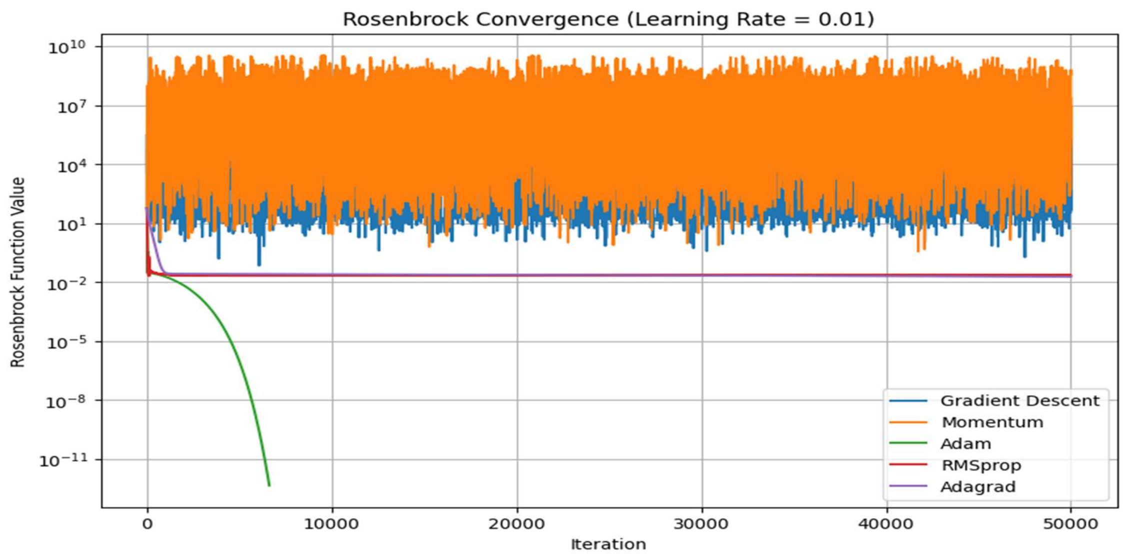
Setting	Value
Learning rates	0.01, 0.05, 0.10
Maximum iterations	50,000
Stopping tolerance	1e-8
Gradient clipping threshold	1000
Initial Rosenbrock point	[0.5, 1]
Initial Ackley point	[4]

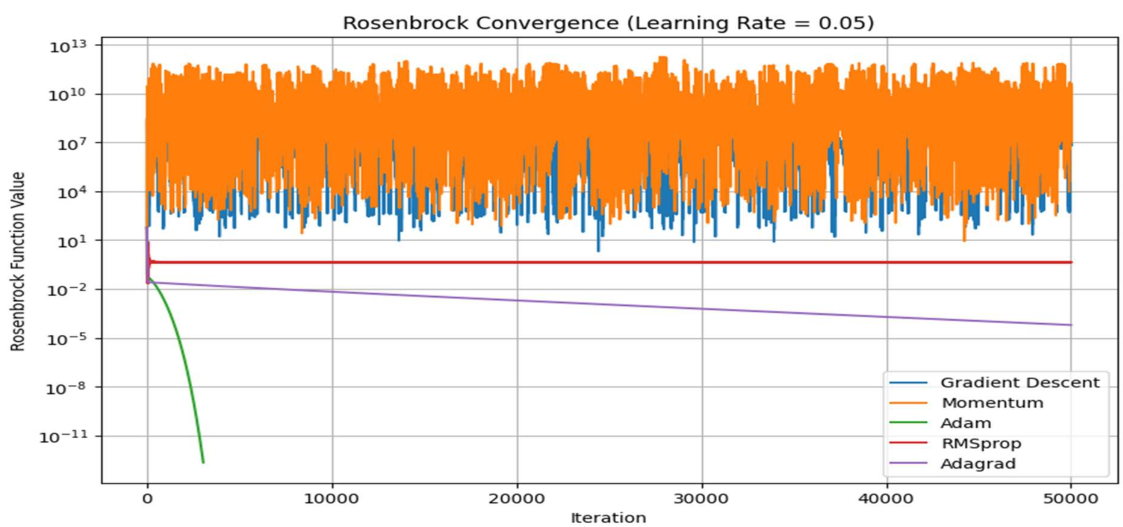
4. **Results**

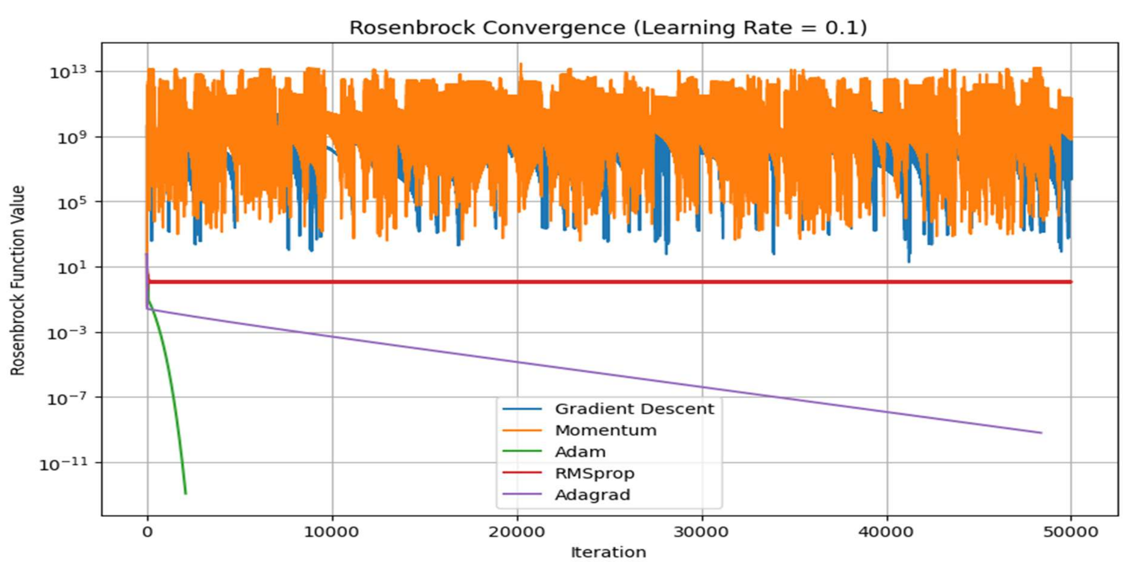
4.1 Rosenbrock Function experiment outcomes on different learning rates

	Learning Rate	Optimizer	Final x	Final y	Final f(x,y)	Iterations	Time (seconds)	Status
0	0.01	Gradient Descent	-6.387857	10.497436	9.190767e+04	50000	2.671795	Maximum iterations
1	0.01	Momentum	28.103848	14.843196	6.006061e+07	50000	2.486498	Maximum iterations
2	0.01	Adam	0.999999	0.999999	4.407535e-13	6625	0.376658	Converged
3	0.01	RMSprop	0.980149	0.975543	2.245074e-02	50000	2.111165	Maximum iterations
4	0.01	Adagrad	0.863607	0.745268	1.863315e-02	50000	1.894554	Maximum iterations
5	0.05	Gradient Descent	-66.676901	528.506268	1.534531e+09	50000	1.824791	Maximum iterations
6	0.05	Momentum	-59.228748	489.803610	9.109815e+08	50000	2.034977	Maximum iterations
7	0.05	Adam	1.000000	0.999999	2.165522e-13	3050	0.148078	Converged
8	0.05	RMSprop	0.675000	0.515625	4.656250e-01	50000	3.386350	Maximum iterations
9	0.05	Adagrad	0.992162	0.984356	6.151895e-05	50000	2.174506	Maximum iterations
10	0.10	Gradient Descent	-30.382577	779.949548	2.050218e+06	50000	1.817085	Maximum iterations
11	0.10	Momentum	221.717617	2871.313438	2.142523e+11	50000	2.080752	Maximum iterations
12	0.10	Adam	1.000000	0.999999	1.181594e-13	2095	0.127680	Converged
13	0.10	RMSprop	0.300000	0.015000	1.052500e+00	50000	2.095906	Maximum iterations
14	0.10	Adagrad	0.999975	0.999949	6.442609e-10	48374	1.826817	Converged

4.2 Rosenbrock Convergence Graphs on Different Learning Rates



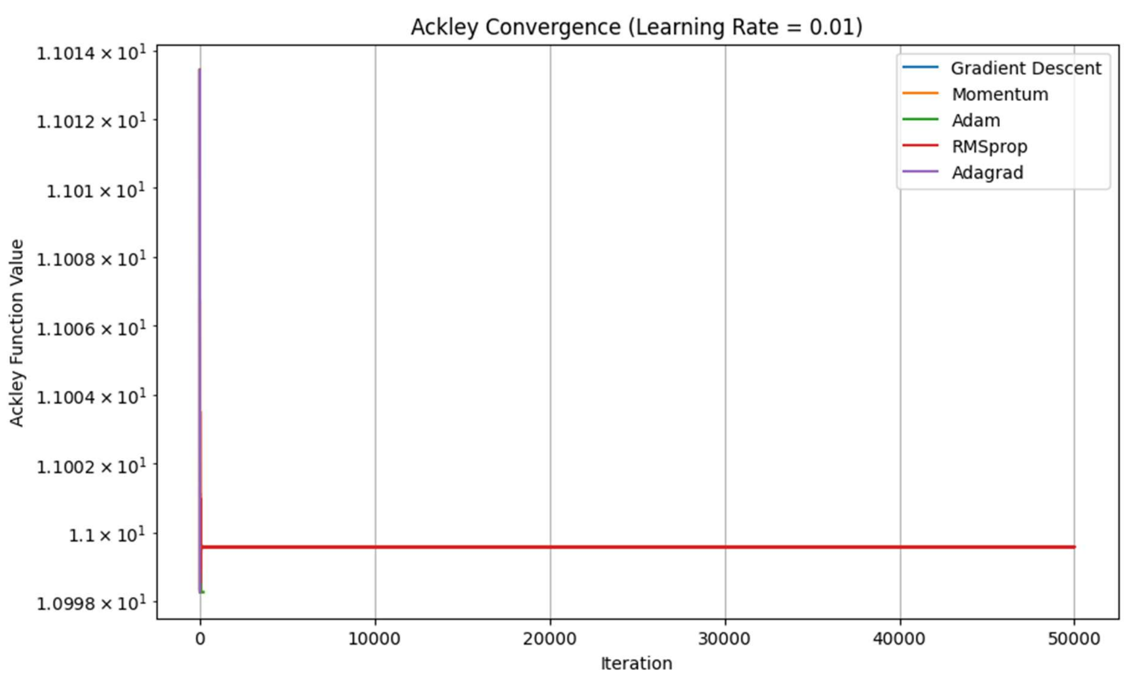


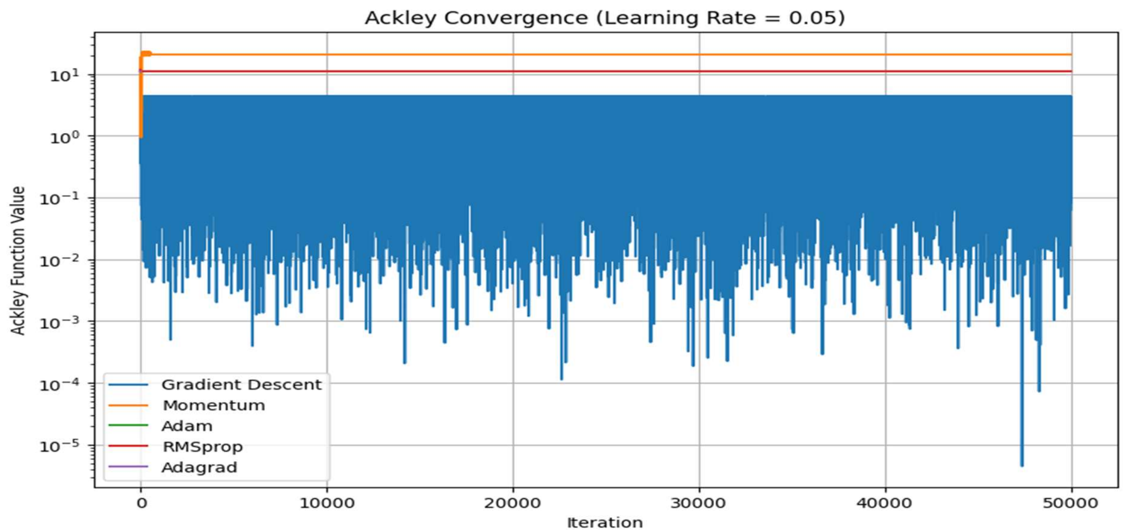


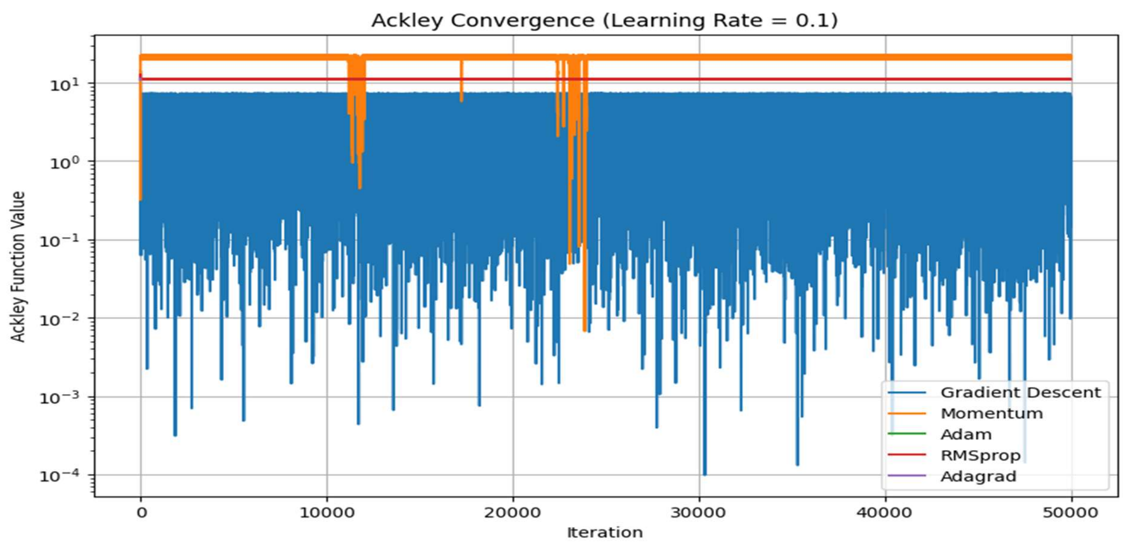
4.3 Ackley Function experiment outcomes on different learning rates

	Learning Rate	Optimizer	Final x	Final f(x)	Iterations	Time (seconds)	Status
0	0.01	Gradient Descent	3.983068	10.998262	6	0.000646	Converged
1	0.01	Momentum	3.983065	10.998262	168	0.015630	Converged
2	0.01	Adam	3.983067	10.998262	209	0.023213	Converged
3	0.01	RMSprop	3.988034	10.999557	50000	3.064596	Maximum iterations
4	0.01	Adagrad	3.983068	10.998262	19	0.001915	Converged
5	0.05	Gradient Descent	0.236179	2.550432	50000	2.171394	Maximum iterations
6	0.05	Momentum	73.883861	20.611076	50000	2.353683	Maximum iterations
7	0.05	Adam	3.983068	10.998262	240	0.026523	Converged
8	0.05	RMSprop	4.007193	11.029113	50000	3.910822	Maximum iterations
9	0.05	Adagrad	3.983068	10.998262	17	0.001979	Converged
10	0.10	Gradient Descent	0.991967	3.602515	50000	2.347533	Maximum iterations
11	0.10	Momentum	555.190338	21.276111	50000	2.560812	Maximum iterations
12	0.10	Adam	3.983067	10.998262	222	0.019531	Converged
13	0.10	RMSprop	3.929165	11.137523	50000	2.679675	Maximum iterations
14	0.10	Adagrad	3.983068	10.998262	25	0.002460	Converged

4.4 Ackley Convergence Graphs on different learning rates







5. <u>**Conclusion**</u>

- **Rosenbrock:** Adam with **learning rate 0.1** was the best-performing configuration.
- **Ackley:** Gradient Descent with **learning rate 0.05** reached the lowest function value, but did not converge within the iteration limit.
- For **stable convergence on Ackley**, **learning rate 0.01** performed better.
- Higher learning rates caused instability for standard Gradient Descent and Momentum, while Adam maintained stable convergence.